Abbreviations

~M 4 = signrtpeQuote(mRPE,Good,hbind(pk(skA 1),hpol(
pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),
exact(s1,exact(prodA),V1,OnlyGood,Assign(P1)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2)))))),akRPE_1)

~M_5 = pk(skA 1)

~M_6 = hpol(pol(P1_Members2,OnlyGood,OnlyGood,Req(
exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2))))

~M_7 = signrtpeQuote(mRPE,Good,hbind(pk(skE_2),hpol(
pol(P2_Members2,Both,Both,Req(exact(mE1),exact(
s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2))))),akRPE_1)

~M_8 = pk(skE_2)

~M_9 = hpol(pol(P2_Members2,Both,Both,Req(exact(mE1),
exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2))))

~X_1 = (P2,~M_7,~M_8,hpol(pol(P2_Members2,Both,Both,Req(
exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2))))
= (P2,signrtpeQuote(
mRPE,Good,hbind(pk(skE_2),hpol(pol(P2_Members2,
Both,Both,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))),akRPE_1),
akRPE_1),pk(skE_1),hpol(pol(P1_Members2,OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)))))

~M_10 = signr(policyMsg(pk(skE_2),pol(P1_Members2,
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))),
skI_1)

~X_2 = (P1,~M_4,~M_5,hpol(pol(P1_Members2,OnlyGood,OnlyGood,
Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2))))
= (P1,signrtpeQuote(
mRPE,Good,hbind(pk(skA 1),hpol(pol(P1_Members2,
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),
akRPE_1),pk(skI_1),hpol(pol(P1_Members2,OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)))))

~M_11 = signr(policyMsg(pk(skI_1),pol(P2,Members2,
Both,Both,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)))),skX_2)

~M_12 = signr(confirmMsg(pk(skZ_2),hstart(mkstart(OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))),skI_1)

~M_13 = signr(confirmMsg(pk(skI_1),hstart(mkstart(OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))),skX_2)

~M_17 = pk(skE_3)

~M_18 = pk(skA 1)

~M_19 = hstart(mkstart(OnlyGood,OnlyGood,Req(exact(
mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(
P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,
OnlyGood,Assign(P2))))

~M_20 = signr(assert(pk(skE_3),pk(skI_1),hstar(mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),
skI_1)

A trace has been found.

Honest Process

Attacker

{1}new akRPE_1
{2}new akTEE_1
{3}new akADV_1

(~M,~M_1,~M_2,~M_3) = (pk(akRPE_1),pk(akTEE_1),
pk(akADV_1),akADV_1)

!

Beginning of process rpe1

{9}new skI_1

{11}event PolicyDeclared(P1,pol(P1,Members2,OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))

Beginning of process rpe2

Beginning of process instance

Beginning of process instance

(P1,~M_4,~M_5,~M_6)

PermissiveRpeTCB

{95}new skZ_2

{97}event PolicyDeclared(P2,pol(P2,Members2,Both,
Both,Req(exact(mE1),exact(s1),exact(prodA),V1,
OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))

(P2,~M_7,~M_8,~M_9)

~X_1

{20}event EvidenceChecked(P1,P2,pk(skZ_2),hpol(
pol(P2,Members2,Both,Both,Req(exact(mE1),exact(
s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)))))

{21}event QuoteCompleted(P1)

{22}event PolicySent(P1,pk(skZ_2),pol(P1,Members2,
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

(P1,P2,~M_10)

~X_2

{106}event EvidenceChecked(P2,P1,pk(skI_1),hpol(
pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),
exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2)))))

{107}event QuoteCompleted(P2)

{108}event PolicySent(P2,pk(skI_1),pol(P2,Members2,
Both,Both,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))

(P2,P1,~M_13)

(P2,P1,~M_11)

{26}event PolicyAuthenticated(P1,P2,pol(P2,Members2,
Both,Both,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))

{29}event PolicyAccepted(P1,P2,pol(P2,Members2,
Both,Both,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))

{30}event PolicyAcceptedKeys(P1,P2,pk(skI_1),pk(
skZ_2),pol(P2,Members2,Both,Both,Req(exact(mE1),
exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2))))

{31}event PoliciesComplete(P1)

{34}event Full(join(P1,mkstart(OnlyGood,OnlyGood,
Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)))))

{35}event ConfirmationSent(P1,P2,hstar(mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P1,P2,~M_12)

(P1,P2,~M_10)

{112}event PolicyAuthenticated(P2,P1,pol(P1,Members2,
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{115}event PolicyAccepted(P2,P1,pol(P1,Members2,
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{116}event PolicyAcceptedKeys(P2,P1,pk(skZ_2),
pk(skI_1),pol(P1,Members2,OnlyGood,OnlyGood,Req(
exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2))))

{117}event PoliciesComplete(P2)

{120}event Full(join(P2,mkstart(OnlyGood,OnlyGood,
Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)))))

{121}event ConfirmationSent(P2,P1,hstar(mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P2,P1,~M_13)

(P2,P1,~M_11)

{39}event ConfirmationAuthenticated(P1,P2,hstar(
mkstart(OnlyGood,OnlyGood,Req(exact(mE1),exact(
s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)))))

{41}event ConfirmationAccepted(P1,P2,hstar(mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P1,P2,~M_12)

{42}event Confirm(P1,mkstart(OnlyGood,OnlyGood,
Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2))))

{125}event ConfirmationAuthenticated(P2,P1,hstar(
mkstart(OnlyGood,OnlyGood,Req(exact(mE1),exact(
s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)))))

{127}event ConfirmationAccepted(P2,P1,hstar(mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

Beginning of process attest

(Identity(mE1,s1,prodA,V1),Good)

{128}event Confirm(P2,mkstart(OnlyGood,OnlyGood,
Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2))))

{179}new skE_3

Beginning of process attest

~M_14 = pk(skE_3)

~M_14 = pk(skE_3)

{48}new challenge_5

{49}event ChallengeSent(P1,pk(skE_3),challenge_5)

~M_15 = challenge_5

~M_15 = challenge_5

{182}event QuoteProduced(pk(skE_3),challenge_5,
Identity(mE1,s1,prodA,V1),Good)

~M_16 = signrtpeQuote(Identity(mE1,s1,prodA,V1),
Good,hchal(challenge_5,pk(skE_3))),akTEE_1)

~M_16 = signrteeQuote(Identity(mE1,s1,prodA,V1),
Good,hchal(challenge_5,pk(skE_3))),akTEE_1)

{52}event QuoteReceived(P1,E1,pk(skE_3),challenge_5)

{54}event QuoteSignatureChecked(P1,E1,pk(skE_3),
challenge_5,Identity(mE1,s1,prodA,V1),Good)

{56}event QuoteVerified(P1,E1,pk(skE_3),challenge_5,
Identity(mE1,s1,prodA,V1),Good)

{80}event QuoteSatisfied(P1,E1,pk(skI_1),pk(skE_3),
challenge_5,Identity(mE1,s1,prodA,V1),Good,mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{88}event CertIssued(P1,E1,pk(skI_1),pk(skE_3),
challenge_5,Identity(mE1,s1,prodA,V1),Good,mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))),
hcert(pk(skE_3),pk(skI_1),hstar(mkstart(OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))))

(~M_17,~M_18,~M_19,~M_20)

(Identity(mE2,s2,prodA,V1),Good)

{179}new skE_4

~M_21 = pk(skE_4)

~M_21 = pk(skE_4)

{134}new challenge_6

{135}event ChallengeSent(P2,pk(skE_4),challenge_6)

~M_22 = challenge_6

~M_22 = challenge_6

{182}event QuoteProduced(pk(skE_4),challenge_6,
Identity(mE2,s2,prodA,V1),Good)

~M_23 = signrtpeQuote(Identity(mE2,s2,prodA,V1),
Good,hchal(challenge_6,pk(skE_4))),akTEE_1)

~M_23 = signrteeQuote(Identity(mE2,s2,prodA,V1),
Good,hchal(challenge_6,pk(skE_4))),akTEE_1)

{138}event QuoteReceived(P2,E2,pk(skE_4),challenge_6)

{140}event QuoteSignatureChecked(P2,E2,pk(skE_4),
challenge_6,Identity(mE2,s2,prodA,V1),Good)

{142}event QuoteVerified(P2,E2,pk(skE_4),challenge_6,
Identity(mE2,s2,prodA,V1),Good)

{171}event QuoteSatisfied(P2,E2,pk(skZ_2),pk(skE_4),
challenge_6,Identity(mE2,s2,prodA,V1),Good,mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{174}event CertIssued(P2,E2,pk(skZ_2),pk(skE_4),
challenge_6,Identity(mE2,s2,prodA,V1),Good,mkstart(
OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))),
hcert(pk(skE_4),pk(skZ_2),hstar(mkstart(OnlyGood,
OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2))))))